AI26501 • MODERN AI ASSIGNMENTS • SP-2026

Robustness of AI-Text Detection Under Text Editing and Paraphrasing

Assignment 16 | Group 16

Extended evaluation: verified light edits, T5 paraphrases, and a length-matched truncation control

Revised 12 October 2026 • Add presenter names before submission

2. Problem and research question

Core question

How much does AI-text detection change when held-out AI passages are edited, paraphrased, or shortened?

Why it matters

False positives can mislabel human writing. False negatives can let AI-origin text pass undetected. A detector score is not proof of authorship.

300
paired topics

5
AI conditions

600
texts per condition

3. Dataset and leakage control

HC3 prepared subset

60,955 training passages
15,157 test passages
17,528 training topics • 4,383 held-out topics

Split design

GroupShuffleSplit
Test size = 0.20 • random_state = 42
Grouped by topic_id
Observed train/test topic overlap = 0

Paired sample: 300 held-out topics. Each condition uses the same 300 human controls plus 300 AI texts. Source distribution is imbalanced; Reddit ELI5 dominates.

4. Conditions and controls

Each condition: 300 human controls + 300 AI passages

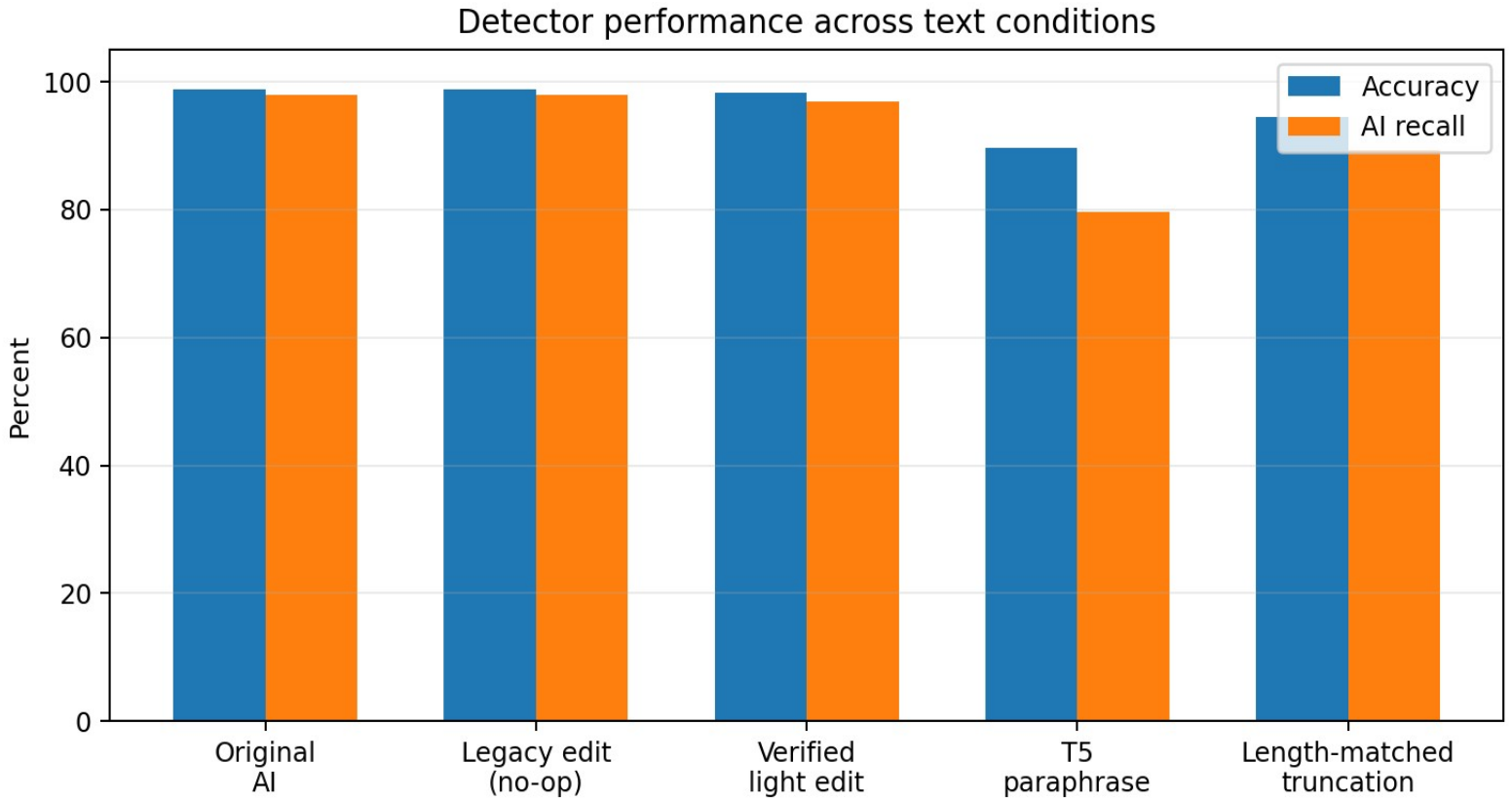
Original AI	Unchanged held-out AI passage
Legacy edit	Phrase-removal function; changed 0/300 passages
Verified light edit	Changed 104/300 passages; mean length ratio 0.999
T5 paraphrase	Generated rewrite; mean length ratio 0.561
Truncation control	Original prefix truncated to paraphrase word count; one shorter original remains unchanged

5. Detector and evaluation protocol

1. Prepare HC3 and group topics
2. Train TF–IDF word uni/bi-grams + LinearSVC on original training text
3. Apply each edit to held-out AI texts only
4. Reuse identical human controls across conditions
5. Report metrics, Wilson CIs, and paired exact McNemar tests

6. Results: overall metrics

Wilson 95% confidence intervals shown in report; paired test results on next slide



AI recall (FPR = 0.33% in every condition)
Original: 98.00%
Verified light edit: 97.00%
T5 paraphrase: 79.67%
Truncation: 89.33%

The legacy edit changed no text and is a no-op diagnostic.

7. Paired changes and statistical tests

Paired comparison	Direction of changes	Exact McNemar p
Original → light edit	3 detected→missed; 0 reverse	p = 0.250
Original → T5 paraphrase	57 detected→missed; 2 reverse	p = 6.14e−15
Original → truncation	26 detected→missed; 0 reverse	p = 2.98e−08
Truncation → T5 paraphrase	38 detected→missed; 9 reverse	p = 2.49e−05

A paired test assesses prediction changes, not whether paraphrases preserve meaning or whether compression explains all effects.

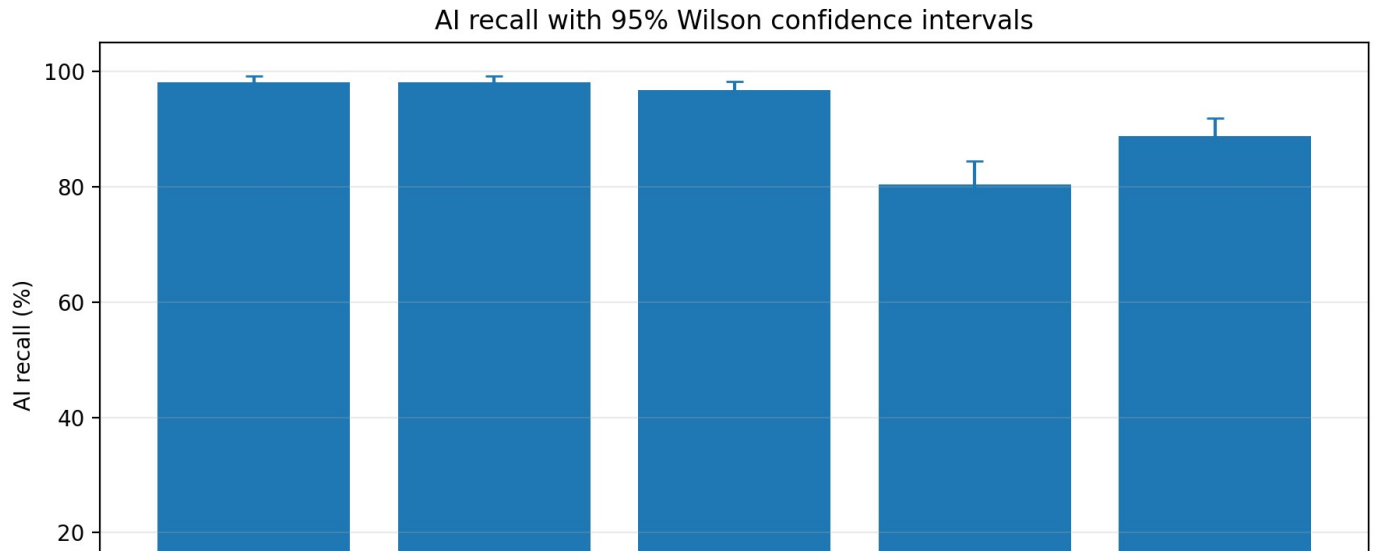
8. Length is a confound

Mean length ratio vs original

T5 paraphrase: 0.561
Truncation control: 0.561
Verified light edit: 0.999

Interpretation

Truncation reduced recall to 89.33%. T5 paraphrases reduced it further to 79.67%. The paraphrase-versus-truncation paired test was significant (p = 2.49e-5), but the experiment does not isolate a pure semantic effect.



9. Supplementary larger-sample light-edit test

All held-out AI passages (n = 4,959)

Original detection: 98.37%
Light-edited detection: 98.02%
17 AI predictions changed
Detection-rate change: −0.34 pp

Human-only held-out group

75/10,198 false positives
FPR = 0.74%
95% Wilson CI ≈ 0.59–0.92%

This supplementary deterministic edit uses a different procedure and sample from the 300-topic paired pilot. Do not merge the estimates.

10. Limitations and conclusion

What the results support

- Verified light edit: 104/300 passages changed; recall 97.00%; p = 0.25.
- T5 paraphrase: recall 79.67%, below original and truncation.
- Length/compression contributes to the observed drop.

What remains uncertain

- Paraphrase meaning preservation
- Generalization beyond HC3 and this detector
- Source imbalance and small subgroups
- Robustness to human edits and other models

Conclusion: report robustness to these specific transformations, not a universal detector capability.

References: HC3 (Guo et al., 2023) • Hugging Face paraphraser model • scikit-learn